

Gene Name	SLC1A1
Minor Allele	C
My Results	GC
My Variants	You have 1 minor allele
Prevalence	44.1% of people have this result
Impact	Medium

"Solute Carrier Family 1 Member 1" gene

Individuals with this SNP are likely prone to greater glutamate levels. Glutamate is a brain chemical (neurotransmitter) that can irritate and overexcite the brain. Glutamate in excess can contribute to OCD and OCD symptoms such as nail biting, skin picking and hair pulling.

- The C allele has been associated with an increased risk for OCD. If combined with the gene variant above, there is a larger increased risk of OCD.
- This variant can increase the risk of treatment-resistant depression.

Diet: Foods that contain vitamin D, like fish and fortified milk, and foods with vitamin B12, such as clams and oysters, may be beneficial. You can ask your clinician for other recommendations for foods containing these vitamins. Eating a diet rich in selenium (for example eating 1-2 brazil nuts per day), can also be beneficial.

Supplements: NAC (N-acetylcysteine) can help lower glutamate. Vitamin D, vitamin B12, and 5HTP may also be of benefit and these supplements can be considered with guidance on dosing, potential risks, and benefits.

Gene Name	COMT V158M
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	50% of people have this result
Impact	Medium

"Catechol-O-Methyltransferase" gene

Individuals with this SNP have a reduced ability to break down dopamine (motivation and reward brain chemical), norepinephrine, and adrenaline (fight or flight brain chemicals). High dopamine can encourage repetitive, reward-related behavior, and has been associated with OCD.

- This variant increases the risk of OCD, more so in men than women. It also increases the risk of anxiety.
- Two variants have been associated with:
 - More intolerance to caffeine.
 - Higher sensitivity to pain after surgery, and greater response to opiates when prescribed for pain.
 - **As a benefit** with motor learning skills.
- COMT is also involved in the breakdown of estrogen (hormone), so those with two copies may also have higher levels of estrogen.

Diet: Foods rich in magnesium, such as pumpkin and soybeans, and zinc, such as oysters and spirulina, may also be beneficial. Avoiding a sugar diet is also recommended.

Supplements: Ashwagandha can help lower cortisol levels. Zinc, magnesium, inositol, curcumin, and L-theanine may also be beneficial and can be considered with guidance on dosing.

Note: If you have two variants you may want to avoid quercetin, as it further reduces your ability to break down these brain chemicals.

Vitamin B6

Vitamin B6 performs a variety of functions and is involved in cognitive development and clearing homocysteine. Elevated homocysteine levels have been associated with neurological conditions such as Alzheimer's Disease, vascular dementia and schizophrenia.

The ALPL gene produces an enzyme that clears vitamin B6. Variants in this gene are associated with lower B6 levels. B6 is used by a gene called CBS to clear homocysteine. Vitamin B6 supplementation has been shown to lower homocysteine. Discuss with your clinician potential interventions that are geared at raising B6 levels through diet and supplementation.

Common symptoms of low vitamin B6 levels include: fatigue, peripheral neuropathy, anemia, sore glossy tongue, mood changes, and cracked and sore lips.

Vitamin B6 Risk SNPs

Gene Name	NBPF3
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	22.9% of people have this result
Impact	High

"Neuroblastoma Breakpoint Family Member 3" gene

With this SNP, you are prone to removing your Vitamin B6 quicker than normal, due to this overactive pathway. This can lead to lower levels of vitamin B6.

- Located close to a gene involved in the clearance of vitamin B6.
- The variant T allele is associated with lower B6 levels.

Supplements: You may benefit from supplementation with Vitamin B6 or the active form P5P.

Diet: You can also increase your intake of dietary Vitamin B6 (see the Complete Vitamin B6 Containing Foods handout)

Note: It is important to remember that we want just the right amount of B6 - not too much or too little.

Gene Name	ALPL
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	30% of people have this result
Impact	High

"Alkaline Phosphatase, Liver/Bone/Kidney" gene

With this SNP, you are prone to removing your Vitamin B6 quicker than normal, due to this overactive pathway. This can lead to lower levels of vitamin B6.

- This protein is involved in the clearance of vitamin B6.
- Variant G allele is associated with lower B6 levels.

Supplements: You may benefit from supplementation with Vitamin B6 or the active form P5P.

Diet: You can also increase your intake of dietary Vitamin B6 (see the Complete Vitamin B6 Containing Foods handout)

Note: It is important to remember that we want just the right amount of B6 - not too much or too little.

Vitamin B12

Vitamin B12 is essential not only for the formation of red blood cells, but also for many other cellular functions. Vitamin B12 is used to clear homocysteine and is important for proper brain and neurological function. Low vitamin B12 is associated with neurological conditions such as Alzheimer's Disease and vascular dementia, and can also contribute to high homocysteine.

Seniors tend to have lower vitamin B12 levels due to problems with acid production and intrinsic factor, which is a molecule important in the absorption of vitamin B12. Vegetarians and vegans are at a high risk for B12 deficiency as almost all naturally occurring B12 is found in meat and dairy.

Cognitive health studies show higher cognitive function and less cognitive loss in individuals in the highest tertile of B12 levels. There are some B12 related variants that make it harder for tissue levels (TCN1 and TCN2) or mitochondrial levels (MUT) to stay normal even in the presence of normal blood B12 levels. Individuals with variants in this pathway may need higher levels of blood B12 in order to maintain adequate brain, mitochondrial and other tissue B12 levels.

Vitamin B12 Risk SNPs

Gene Name	TCN2
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	19.8% of people have this result
Impact	High

"Transcobalamin 2" gene

Individuals with 2 copies of this SNP are prone to significantly lower levels of B12 in the brain. The carrier protein encoded by this gene only binds 10-20% of total B12, but is responsible for carrying 80-90% of the B12 into the brain. Individuals with this SNP may require higher levels of B12 in the blood to have adequate levels of B12 in the brain.

- This protein binds vitamin B12 and transports it into cells.
- GG variants have been associated with less efficient binding and transport of B12, correlating with reduced tissue levels of vitamin B12 (holotranscobalamin) even when B12 levels in the blood appear normal.
- GG variants have been associated with 3.33 times the risk of peripheral neuropathy even with normal B12 levels. Even greater risk (6.9 times) if folate intake is >800 mg/day.
- G allele has been associated with 1.51 times the risk for endometriosis.
- C allele (1 variant) has been shown to convey relative protection again Alzheimer's type dementia when combined with TT genotype of rs1801131 (MTHFR A129C).
- Individuals with this variant may need higher than average B12 levels in order to have adequate brain and tissue levels.

Supplements: Supplementing Vitamin B12 via injections, sublingual or oral forms can be beneficial to maintain adequate B12 levels. You may need higher doses to get enough B12 to the brain.

Diet: Increasing dietary intake of Vitamin B12 from foods such as meats, poultry, fish, etc. Ask your clinician for the Vitamin B12 Containing Foods handout.

Gene Name	GSTP1
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	11.1% of people have this result
Impact	High

"Glutathione S-Transferase Pi 1" gene

Glutathione is an enzyme that is important for the removal of mercury and other toxins, as well as chemicals from the body. This variant in the glutathione-related gene (GSTM1) results in a lower ability to get rid of heavy metals, pesticides, and other toxins such as smoke by-products. This can lead to high mercury and other toxin levels. Lower ability to detox can contribute to a greater risk of cognitive decline, birth defects, and other health conditions when exposed to glyphosates (RoundUp), pesticides, and other toxins.

- Glutathione is the master antioxidant for the body, important for detoxification.
- Minor allele G is associated with elevated mercury levels and excess susceptibility to the detrimental cognitive effects of cumulative lead exposure.
- Minor allele G is associated with difficulties removing glyphosate (RoundUp) from the body. The study showed G allele had a 2.6 times the risk for miscarriages and birth defects in infants born to mothers with VERY high glyphosate exposure. If two variants (GG) 4.8 times the risk.
- Two variants (GG) have been associated with nearly 50% decreased ability to clear smoke by-products from the body and a 1.84 times the risk for cognitive impairment/ Alzheimers.
- GSTP1 plays an important role in detoxifying carcinogens that can lead to cancer risks.
- Involved in over 200 different diseases.

Diet: Getting optimal amounts of Vitamin C, via dietary options, and avoiding glyphosate exposure from non-organic sources. See the "Clean 15 and Dirty Dozen" and "Glyphosate - Foods & clearing toxins" handout.

Supplements: Supplementing with products such as sulforaphane, N-acetylcysteine (NAC), direct glutathione replacement, and vitamin C can help to support this pathway and your ability to remove toxins from the body.

Gene Name	NFE2L2
Minor Allele	T
My Results	TG
My Variants	You have 1 minor allele
Prevalence	21.4% of people have this result
Impact	Medium

"Nuclear Factor Erythroid 2-Related Factor 2" gene

With this variant, you may have less ability to turn on various detox pathways, such as many of the glutathione pathways above. This can lead to increased oxidative stress (i.e. more "rusting" or "tarnishing" of the cells) and higher susceptibility to the damaging effects of different toxic chemicals and substances.

- Codes for the transcription factor Nrf2 (master regulator of detoxification/ inflammation - acting as a switch), which turn on many other genes, which are important for reducing oxidative stress.
- Variants have been associated with decreased Nrf2 expression.
- May be at greater risk for cerebrovascular disease (a disease that affects the blood vessels and blood supply to the brain).
- Two variants (TT) convey a lower risk for hypertension or elevated blood pressure.

Supplements: Sulforaphane, hydrogen water, resveratrol and cordyseps can be beneficial options to support this variant.

Note: Lowering inflammation, in general, can support this pathway as well.

Gene Name	ABCC1
Minor Allele	C
My Results	CT
My Variants	You have 1 minor allele
Prevalence	48.7% of people have this result
Impact	Medium

"ATP Binding Cassette Subfamily C Member 1" gene

This gene is important for the ability to transport mercury, mold-related toxins, and other toxins out of the brain. With this variant, you may have less of an ability to remove these toxins from the brain and can result in a greater risk of mercury toxicity in the brain. Additionally, this variant can increase the risk for neurodevelopmental disorders in children born to mothers consuming high amounts of high mercury foods, and may also influence mold-related cognitive issues.

- Involved in transport and elimination of drugs and toxins such as mercury.
- Carriers of the C allele less able to remove mercury from the brain and nervous tissue.
- Individuals with two variants (CC) has been associated with:
 - Higher mercury **tissue** levels with higher fish intake than individuals with TT variants.
 - Lower neurodevelopmental scores in children born to mothers who consumed a high fish diet. A particular type of mercury (methyl mercury) easily crosses the blood-placenta barrier and accumulated more in the fetus than in the mother.

Diet: You may find benefit from limiting the intake of high mercury fish king mackerel, shark, tilefish, swordfish, bigeye tuna, marlin, orange roughy etc.), particularly if 2 variants (AA).

Supplements: Sulforaphane, NAC (particularly if high mercury exposure from fish) may be beneficial to support this pathway.

Gene Name	GSTM1
Minor Allele	A
My Results	A
My Variants	You have 2 minor alleles
Prevalence	50.9% of people have this result
Impact	High

"Glutathione S-Transferase Mu 1" gene

Glutathione is an enzyme that is important for the removal of mercury and other toxins, as well as chemicals from the body. You are completely missing this glutathione-related gene (GSTM1) due to this deletion. Decreased ability to clear these toxins can contribute to many health conditions including the risk of certain cancers, Alzheimer's, hardening of the arteries (atherosclerosis), and asthma.

- Important enzyme in the glutathione detox system. Glutathione is the master antioxidant for the body, important for detoxification.
- Individuals with the "A" variant have a complete deletion of this gene and it is associated with increased oxidative stress on cells and lower ability to clear toxins (mercury, glyphosate/pesticides) from cells.
- Also associated with risk of Alzheimer's, atherosclerosis, asthma, and lung cancer development as well as low birth weight for babies of women with this SNP.
- The combination of multiple variants in the other glutathione-related genes can have an additive effect for the risk of conditions such as lung cancer.

Diet: Getting optimal amounts of Vitamin C, via dietary options, and avoiding glyphosate exposure from non-organic sources. See the "Clean 15 and Dirty Dozen" and "Glyphosate - Foods Clearing Toxins" handout.

Supplements: Supplementing with products such as sulforaphane, N-acetylcysteine (NAC), direct glutathione replacement, and vitamin C can help to support this pathway and your ability to remove toxins from the body.

Anxiety and Stress Vulnerability

Anxiety can manifest in many different ways. Some general symptoms of anxiety include persistent and excessive worrying, repetition of words or phrases, or fixating on worrying. Symptoms can include episodes of panic. Panic can manifest with physical symptoms such as rapid heart rate, sweating, along with doom-type symptoms or fears. More extreme forms of anxiety manifest as phobias (an extreme or irrational fear). Generalized Anxiety Disorder (GAD), panic disorder, agoraphobia (fear of leaving one's home or crowded places), social phobia, and specific phobias can have some overlapping and some very different contributing factors.

The genomics of anxiety are complex, but this panel looks at small changes in DNA called SNP's or variants that relate to brain chemicals like serotonin ("happy and calming" brain chemical), norepinephrine/adrenaline (involved in the "fight or flight" response), dopamine (the "feel good" brain chemical), cortisol (the "stress" hormone), and more. We want the right balance of these brain chemicals. Variants can result in abnormalities in these brain chemical levels, expression, and activation, resulting in greater risk for anxiety and anxiety-like symptoms. Anxiety disorders are prevalent in the world, and it has been estimated that approximately one-third of people will develop anxiety at some point in their lives. Genomics can provide insight into one's risk factors as well as personalized interventions to assist you.

Anxiety and Stress Vulnerability Risk SNPs

Gene Name	ADRA1A
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	6.5% of people have this result
Impact	High

"Alpha-1-adrenergic Receptor" gene

Individuals with this SNP may have an increased adrenaline (fight or flight hormone) response which can increase the risk for anxiety, especially in older patients.

- In individuals aged 65 and over, 2 variants have been associated with over 4 times the risk for generalized anxiety disorder (GAD).
- You may have an increased risk of ADHD.

Diet: Maintaining a low carbohydrate and low sugar diet may help. Incorporating magnesium into your diet may reduce anxiety levels.

Lifestyle: Meditation or yoga can be beneficial with this variant.

Supplements: Magnesium, vitamin B6, ashwagandha, L-theanine, or magnolia may help decrease your adrenaline response and can be used with guidance for proper dosing.

Gene Name	RGS2
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	7.8% of people have this result
Impact	High

"Regulator of G Protein Signaling 2" gene

Serotonin is the happy, calm brain chemical. You may make less of a protein that is important for serotonin release. This leads to less serotonin release from your neurons and lower serotonin signaling in parts of the brain that regulate anxiety.

- This is associated with a slightly increased risk for panic disorder as well as an increased risk of generalized abdominal pain.
- You may have a 50% less chance of responding to sertraline (antidepressant/SSRI medication) treatment in patients with anxiety.

Diet: Eating a diet rich in tryptophan may be beneficial. Tryptophan-rich foods include turkey, chicken, dairy, fish, egg white, pumpkin seeds, and peanuts.

Supplements: Extended-release 5-HTP, and forskolin have evidence in helping improve this pathway and can be considered with guidance on dosing, risks, and potential benefits.

Medications: Oxytocin may be beneficial. You may be less likely to benefit from SSRIs (fluoxetine, escitalopram, sertraline, etc.) than typical. Your clinician may also want you to avoid the medication Xyrem (commonly prescribed for narcolepsy) as it lowers the activity of this protein even more.

Gene Name	ESR2
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	12.2% of people have this result
Impact	High

"Estrogen Receptor Beta" gene

Stimulation of estrogen receptors can help decrease anxiety levels. You may have less estrogen receptor activity which can lead to anxiety and anxiety-like symptoms.

- In women over 65 years of age, this variant has been associated with double the risk for generalized anxiety disorder (GAD).
- For women not on hormone replacement, this variant may also increase the risk of depression.

Supplements: Genistein, resveratrol (in red grapes), pterostilbene (in blueberries), and rhubarb extract (ERr 731) can help improve estrogen levels but should be used with guidance for proper dosing, risks, and potential benefits.

Medications: Progesterone supplementation can make anxiety worse, so if you are considering hormone replacement, you may want to let your clinician know.

Gene Name	ADRA1A
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	16% of people have this result
Impact	High

"Alpha-1-adrenergic receptor" gene

Individuals with this SNP may have an increased adrenaline (fight or flight hormone) response which can increase the risk for anxiety, especially in elderly patients.

- A study showed that people over the age of 65 have 4.48 times increased risk of generalized anxiety disorder (GAD).
- You may have an increased risk of ADHD.

Diet: Maintaining a low carbohydrate and sugar diet may help.

Incorporating magnesium into your diet may also be beneficial in reducing anxiety levels.

Lifestyle: Meditation or yoga can be beneficial with this variant.

Supplements: Magnesium, vitamin B6, ashwagandha, L-theanine, or magnolia may help decrease your adrenaline response and can be used with guidance for proper dosing.

Gene Name	NPSR1
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	19.6% of people have this result
Impact	High

"Neuropeptide S Receptor 1" gene

Individuals with this SNP tend to have overactive "wakefulness" in the brain due to high NPSR1 receptor activity. This can induce anxiety and panic, increased flushing response, and increased histamine release during times of stress.

- Individuals that have had childhood trauma AND who have 2 variants are more sensitive to anxiety.
- This variant can lead to you over-interpreting reactions to fear, as well as increase your risk of having difficulty sleeping when you are stressed.

Lifestyle: You may want to limit caffeine use as it can increase orexin (a brain chemical that this variant may increase) even more.

Supplements: Pycnogenol, quercetin, bromelain, vitamin C, melatonin, and luteolin may be beneficial at helping decrease histamine release and/or "hyperarousal", but should be used with guidance regarding dosing, risks, and potential benefits.

Medications: Belsomra has evidence to decrease orexin (the brain chemical that this variant may increase) if you have insomnia.

Obsession, Rumination, and Compulsions

Obsessive Compulsive Disorder (OCD) is a common, complex psychiatric disorder that is characterized by uncontrollable, recurring behaviors, known as "compulsions", and recurring thoughts, known as "obsessions". These traits can interfere with everyday life, work, school, and relationships.

There is growing evidence that some of the most significant risks in regard to OCD is genetic factors, particularly those that affect neurotransmitter levels in the brain. Neurotransmitters are the brain's chemical messengers responsible for sending signals between nerve cells in the brain. You will notice a lot of discussion on 4 brain chemicals: norepinephrine, dopamine, serotonin, and glutamate in this panel. Norepinephrine plays a role in energy, focus and motivation, dopamine plays a role in pleasure, movement and irritability and serotonin plays a role in mood, happiness and a sense of wellbeing. Glutamate, being the most abundant excitatory neurotransmitter, plays a role in learning, memory and cognition. With these chemical messengers the goal is the right balance - not too much and not too little while having an appropriate amount of each.

These neurotransmitters play a large role in the development of OCD, and can even cause resistance to OCD treatment. Lower serotonin levels, which can be due to genetic or dietary factors, such as low vitamin D, vitamin B12, and high homocysteine, have been correlated with OCD. Higher dopamine and glutamate levels, which are mainly due to genetic factors, have also shown association with OCD. Your licensed health care provider will discuss these variants with you and help you decide if you would benefit from any potential interventions and can help you determine which supplements and dietary changes make sense for you or your child. They also will determine appropriate dosing.

Obsession, Rumination, and Compulsions Risk SNPs

Gene Name	HTR2A
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	16.8% of people have this result
Impact	High

"5-Hydroxytryptamine Receptor 2A" gene

Individuals with this SNP are prone to less effective serotonin (happy, calming brain chemical) signaling. Low postsynaptic (receiving neuron) serotonin signaling is a major contributor to OCD.

- This variant leads to lower levels of serotonin.
- In addition to increasing the risk of OCD, this variant also increases the risk of "tic" disorder, and post-traumatic stress disorder (PTSD).

Diet: Foods that contain vitamin D (fish and fortified milk), foods that contain vitamin B6 (pistachios and chickpeas), foods that contain folate, (spinach and asparagus), and foods with vitamin B12 (clams and oysters) may be beneficial. You can ask your clinician for other recommendations for foods containing these vitamins. Eating a diet rich in selenium (for example eating 1-2 brazil nuts per day), can also be beneficial.

Supplements: Vitamin D, vitamin B6, vitamin B12, and folate are vitamins that may be beneficial. In addition, 5-HTP, L-theanine, certain probiotics, and myoinositol can be considered with guidance on dosing, potential risks, and benefits.

Medications: SSRIs, SNRIs, and TCAs are medications that help raise serotonin levels and can be considered by you and your clinician for treating OCD.